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# fp-tools: A Reproducible Command-Line Platform for Classical ATAC-seq Footprinting, Motif Discovery Preparation, Pseudobulk Aggregation, and Replicate-Aware Reporting

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## Abstract

(1) Background: Transcription-factor (TF) footprinting from ATAC-seq is central to inferring regulatory activity, but the surrounding ecosystem is fragmented across single-purpose tools with heterogeneous interfaces, inconsistent reproducibility, and uneven test coverage, which raises the cost of building end-to-end regulatory-genomics analyses. (2) Methods: We present `fp-tools`, a standalone, pip-installable Python package that unifies bias correction, footprint scoring, differential TF-binding detection, and aggregate visualization in a reproducible command-line package. Around this core we add optional utilities for de novo motif-discovery preparation, single-cell pseudobulk aggregation, and replicate-aware differential-binding reporting. (3) Results: The package provides deterministic, regression-tested workflows, reproducible figure generators, and a tiered public-data benchmark design for the supported current workflow. Existing classical analyses remain unchanged when optional utilities are not invoked. (4) Conclusions: `fp-tools` integrates established footprinting strengths with pragmatic replicate-aware, pseudobulk, and motif-discovery helper utilities in a single reproducible package, lowering the barrier to standardized, benchmarkable ATAC-seq regulatory analysis.

**Keywords:** ATAC-seq; transcription factor footprinting; chromatin accessibility; motif analysis; pseudobulk ATAC-seq; replicate-aware reporting; reproducibility; command-line bioinformatics

## 1. Introduction

Chromatin accessibility profiling by the assay for transposase-accessible chromatin using sequencing (ATAC-seq) has become a standard readout of the active regulatory landscape of a cell [1]. Within accessible regions, sequence-specific transcription factors (TFs) protect the DNA they occupy from transposition, leaving characteristic depletions—"footprints"—flanked by hyper-accessible edges. Detecting and interpreting these footprints supports inference of TF occupancy, differential regulation between conditions, and the functional consequences of non-coding variation.

A mature but *fragmented* ecosystem of methods addresses pieces of this problem. Classical footprinting and differential-binding frameworks such as TOBIAS [2] and HINT-ATAC [3] correct enzymatic sequence bias and score footprints at a single TF-relevant scale. Multiscale and nucleosome-aware approaches, exemplified by the PRINT/scPrinter family [4], model accessibility structure across a range of footprint widths. Supervised TF-binding-site (TFBS) predictors such as maxATAC [5] and sequence-to-profile models

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such as ChromBPNet [6] learn occupancy from labeled data or raw sequence, and downstream interpretation tools and variant scorers translate models into motif and allele-level effects. Each tool typically ships its own interface, input conventions, and assumptions, so assembling a coherent, reproducible pipeline—and *benchmarking* alternatives on common ground—remains laborious and error-prone.

Beyond this tooling friction, footprint interpretation has practical limitations that make reproducibility and careful signal handling important. Classical footprinting depends on base-resolution cut-site depletion and can be confounded by enzyme sequence bias, local accessibility, motif strength, and factor-specific binding kinetics [7]. Deep sequence-to-profile models and multiscale methods can improve sensitivity for some questions, but they often require substantial compute and specialized training workflows. For many applied ATAC-seq projects, the immediate need is therefore a transparent, repeatable workflow that keeps the classical analysis reliable while exposing selected extensions without forcing users into a different software ecosystem.

Four gaps motivate this work. First, **signal clarity**: users need to know which signal representation is appropriate for footprint scoring and how optional normalization changes interpretation. Second, **interface fragmentation**: moving from bias correction to footprint scoring to differential binding to visualization commonly spans several packages with incompatible I/O. Third, **reproducibility and testing**: many footprinting workflows lack deterministic regression tests, pinned environments, and machine-checkable output contracts, making it hard to know whether a change altered results. Fourth, **extensibility without regression**: adding new scientific behavior, such as multiscale scoring or pseudobulk aggregation, should not destabilize the classical workflow.

We address these gaps with `fp-tools`, a reproducible ATAC-seq footprinting platform that keeps the classical workflow intact while adding a small set of practical optional utilities. The package unifies bias correction, footprint scoring, motif-aware differential binding, and aggregate visualization, then extends this core with de novo motif-discovery preparation, pseudobulk ATAC aggregation, and replicate-aware reporting. The goal is not to replace specialized deep-learning or variant-scoring systems in this first version, but to make a complete footprinting analysis easier to run, audit, repeat, and benchmark. This paper describes the supported methods, tiered validation design, and reproducibility engineering that tie these workflows together.

## 2. Materials and Methods

### *2.1. Package Architecture and Core Workflows*

`fp-tools` is distributed as a pip-installable package (`fp-tools-bio`) with vendored Cython internals for performance-critical signal operations. The core workflow has four method layers: Tn5 sequence-bias correction, footprint scoring over corrected signal, motif-anchored differential TF-binding analysis, and aggregate visualization around TFBS or region sets. These layers share common region, signal, motif, and sample-label conventions so intermediate files can be passed through a complete analysis without format-specific glue code. YAML configuration files provide a reproducible way to save and rerun the same analysis without changing the underlying methods.

For classical footprint scoring, let $x_i$ denote corrected Tn5 cut-site signal around candidate position $p$. The default footprint score is a center-versus-flank contrast,

$$F(p) = \frac{1}{|L| + |R|}\left(\sum_{i \in L} x_i + \sum_{i \in R} x_i\right) - \frac{1}{|C|}\sum_{i \in C} x_i, \tag{78}$$

where $C$ is the protected center and $L$ and $R$ are flanking windows. Larger values therefore correspond to the expected footprint pattern: fewer cuts at the putative bound site and more cuts in nearby accessible flanks.

### 2.2. Optional Utilities

New behavior is added as explicit optional utilities, each inert unless selected:

- *De novo motif-discovery preparation* exports candidate-centered sequences, records reproducible MEME/DREME [8] and Tomtom [9] settings, and summarizes discovered motifs and matches to known motif databases. This can be run as a standalone discovery step or as a supplement to known-motif analyses when a database, such as a JASPAR release, may not contain all active motifs in the sample.
- *Pseudobulk aggregation* groups single-cell ATAC fragments by annotation columns into bulk-like fragment files, manifests, optional indexed outputs, and optional normalized cut-site tracks for downstream footprinting and aggregate plots.
- *Replicate-aware differential binding* treats repeated samples from the same condition as biological replicates, reports condition means and replicate variation, and produces diagnostic tables and figures for differential TF-binding comparisons (Section 2.3).

Conceptually, the optional utilities answer focused questions around the classical core. De novo motif preparation asks which sequence patterns recur among exported candidate sites; replicate-aware reporting asks whether repeated-condition inputs can be summarized as condition means and replicate variation; and pseudobulk aggregation asks whether single-cell fragments can be grouped into interpretable group-level profiles that run through the same footprinting machinery. In this work we validate that path with a public 10x Genomics PBMC Multiome dataset processed by Cell Ranger ARC 2.0.0 [10]. Official 10x gene-expression graph clusters are collapsed into broad PBMC labels using canonical marker-gene enrichments (B cells, CD4 T cells, NK/cytotoxic T cells, CD14 monocytes, FCGR3A monocytes, dendritic cells, and mixed myeloid cells), then ATAC fragments are split by label and converted to CPM-normalized cut-site bigWigs. Motif-associated peak sets for PAX5, TCF7/TCF7L2, CEBPB, and CTCF are taken from the 10x TF-analysis peak–motif mapping distributed with the same dataset.

### 2.3. Replicate-aware Differential Binding and Quantile Normalization

Replicate-aware analysis is integrated directly into `detect-tf-binding`. If users pass repeated condition names, for example `-cond-names Bcell Bcell Tcell Tcell`, fp-tools treats the corresponding signal files as biological replicates of the same condition while preserving the original differential-binding output columns. Let $x_{c,r,i}$ be the footprint score at site or background position $i$ for replicate $r$ of condition $c$. With `-normalization sample-quantile`, each replicate is normalized before condition averaging using the same multiplicative quantile curve used throughout fp-tools. For sample $s$, fp-tools computes empirical positive score quantiles $q_s(p)$ and the reference quantile $q_*(p) = S^{-1}\sum_s q_s(p)$ across the $S$ input samples, then fits a smooth multiplicative factor $g_s(x)$ to the ratio $q_*(p)/q_s(p)$. Each score is transformed as

$$\tilde{x}_{s,i} = \max\{0,\, x_{s,i} g_s(x_{s,i})\}.$$

The default `condition-quantile` mode preserves the previous behavior by first averaging replicate background distributions within each condition, fitting one factor $g_c(x)$ per condition, and applying that condition-level factor to all replicates in the condition before final condition means and SDs are reported; `none` disables this step. The same shared normalization helper and the same sample-first versus condition-first ordering are used by `plot-aggregate`, so aggregate visualizations and `detect-tf-binding` comparisons are normalized consistently. Because this step removes distribution-level depth and background offsets, differential interpretation is based on normalized per-site contrasts and the differential-binding comparison statistic, not on global baseline height alone.

After normalization, condition-level scores are summarized as

$$\bar{x}_{c,i} = \frac{1}{n_c}\sum_{r=1}^{n_c} \tilde{x}_{c,r,i}, \qquad s_{c,i} = \sqrt{\frac{1}{n_c-1}\sum_{r=1}^{n_c}(\tilde{x}_{c,r,i} - \bar{x}_{c,i})^2}.$$

For a comparison between conditions $c_1$ and $c_2$, fp-tools reports both an interpretable footprint-score difference and a stabilized log fold change,

$$\Delta_i = \bar{x}_{c_1,i} - \bar{x}_{c_2,i}, \qquad L_i = \log_2 \frac{\bar{x}_{c_1,i} + \epsilon}{\bar{x}_{c_2,i} + \epsilon},$$

where $\epsilon$ is the pseudocount estimated from the background signal unless supplied explicitly. When both conditions have replicate support, approximate standard errors are reported as

$$\mathrm{SE}(\Delta) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}},$$

$$\mathrm{SE}(L) \approx \frac{1}{\ln 2}\sqrt{\frac{s_1^2}{n_1(\mu_1+\epsilon)^2} + \frac{s_2^2}{n_2(\mu_2+\epsilon)^2}}.$$

The standard `change` and `pvalue` columns remain unchanged for compatibility. Additional columns include replicate counts, condition score SDs, mean $\Delta$, mean $L$, and standard-error summaries. A legacy replicate-report command remains available as a post-hoc helper for existing `*_results.txt` files.

### *2.4. Public Data, Benchmark Tiers, and Metrics*

Benchmarks are organized as versioned TSV manifests recording source, benchmark tier, cell type, donor, TF, assay, accessions, assembly, output type, format, URL, checksum, status, local path, split, and notes. A discovery script queries the ENCODE REST API for released human GRCh38 ATAC-seq and matched TF ChIP-seq/CUT&RUN; a resumable downloader writes per-benchmark download reports with checksums and local paths. Raw and processed public data, and large benchmark outputs, are kept out of version control; scripts, manifests, schemas, tiny fixtures, and figure code are tracked.

The benchmark is deliberately *tiered* rather than monolithic (Table 1): a fast CI-smoke tier on tiny fixtures; a bulk matched tier (ENCODE bulk ATAC vs. matched TF ChIP/CUT&RUN) as the primary classical benchmark; a depth-robustness tier over downsampled ATAC; a de novo motif-preparation tier; and a real-data pseudobulk demonstration tier. For binary TF-binding labels we compute AUROC and AUPRC globally and per TF-cell task, recall at 1%, 5%, and 10% FDR, precision at fixed recall, calibration curves with expected and maximum calibration error, Brier score for probability models, per-TF macro averages, Wilcoxon signed-rank tests for paired method comparisons, and stratified bootstrap confidence intervals for key deltas. Expected calibration error partitions predictions into $B$ equal-width probability bins and averages the gap between confidence and accuracy,

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{n}\left|\mathrm{acc}(b) - \mathrm{conf}(b)\right|,$$

with maximum calibration error defined analogously as the worst-bin gap. Metric, calibration, and bootstrap computations are implemented as standalone, unit-tested scripts and combined by a benchmark pipeline runner that also emits PDF/SVG/PNG figure panels.

**Table 1.** Tiered benchmark design. Large data and outputs are untracked; only manifests, scripts, and small smoke results are committed.

| Tier | Purpose | Primary outputs |
|---|---|---|
| CI smoke | Install/regression checks on tiny fixtures | command success, output existence, stable summaries |
| Bulk matched | Main classical benchmark (ATAC vs. matched TF ChIP/CUT&RUN) | TFBS labels, predictions, AUROC/AUPRC, reports |
| Depth robustness | Shallow-depth sensitivity (50/25/10/5%) | performance-vs-depth and calibration curves |
| De novo motif prep | Candidate FASTA and motif-discovery helper validation | exported FASTA, command plans, motif summaries |
| Pseudobulk | Single-cell extension and real-data demonstration | 10x PBMC cell-type pseudobulks, TF aggregate profiles |

### *2.5. Reproducibility Engineering*

The package ships a unit-test suite covering numerical helpers, CLI behavior, and deterministic golden CLI regressions for footprint scoring, differential binding, and aggregate plotting, with an opt-in slow regression for bias correction. A continuous-integration workflow runs the suite on a clean checkout using only small fixtures. Every benchmark result is expected to store random seeds, tool versions, command lines, and the resolved manifest. Figures are saved as vector PDF/SVG plus 300-dpi PNG previews, with source plotting tables retained for auditability.

## 3. Results

### *3.1. A Unified, Reproducible Footprinting Platform*

`fp-tools` consolidates the four core ATAC footprinting workflows behind a single, consistent workflow with shared region/signal conventions, so a complete analysis—bias correction, footprint scoring, differential binding, and aggregate visualization—runs without crossing package boundaries or reformatting intermediates. Optional YAML configuration records the same analysis in a machine-readable form for reruns and batch processing (Figure 1).

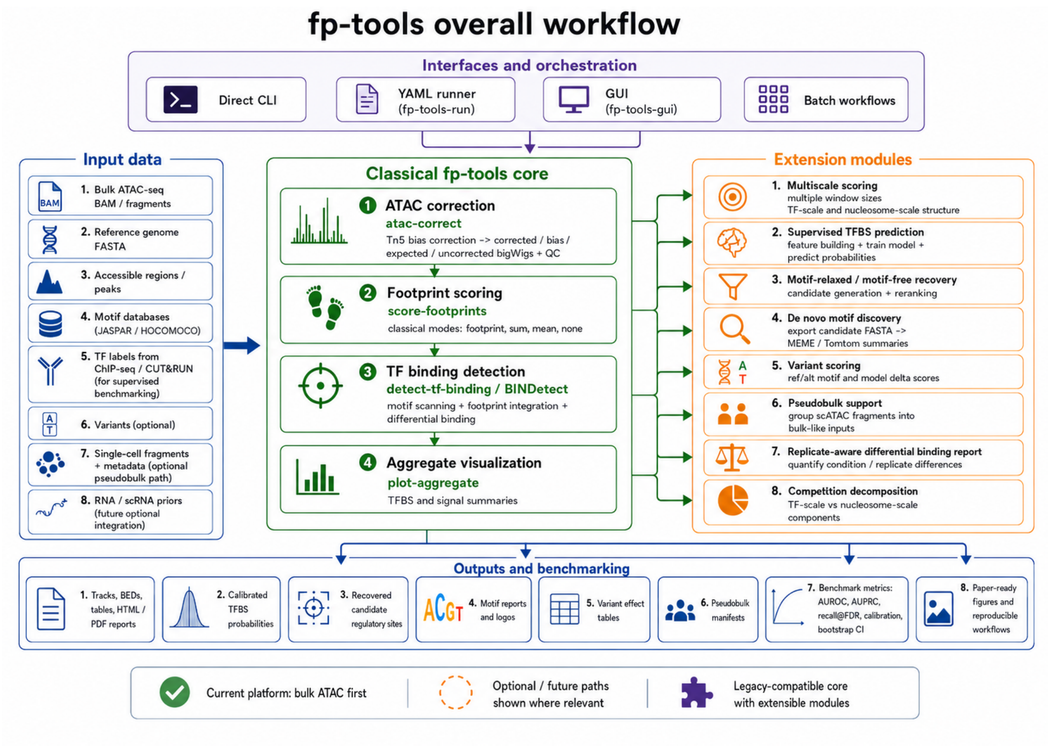


**Figure 1.** Overall `fp-tools` workflow. Bulk ATAC-seq inputs enter a legacy-compatible classical core for Tn5 bias correction, footprint scoring, TF-binding detection, and aggregate visualization. Optional YAML and batch layers record reproducible analyses, while first-version utilities add multiscale scoring, de novo motif preparation, pseudobulk support, and replicate-aware reporting. Outputs include signal tracks, reports, motif summaries, pseudobulk manifests, benchmark metrics, and reproducible paper-ready figures.

### *3.2. Deterministic Regression and CLI Coverage*

The classical workflow is covered by deterministic golden regressions that pin output summaries for footprint scoring, differential binding, and aggregate plotting, plus smoke tests for public entry points and YAML dry-runs. This converts "did this change alter results?" from a manual judgment into an automatically checked contract, and provides the stable base on which the opt-in modules are layered without disturbing default behavior.

### *3.3. De Novo Motif Preparation*

The de novo motif-preparation path exports candidate-centered FASTA files, constructs reproducible MEME/DREME and Tomtom analysis records, and summarizes motif results into TSV/HTML reports with inline consensus logos. This keeps external motif discovery reproducible without making a new motif-calling algorithm part of the first-version claim. The workflow can be used alone to discover recurring patterns in candidate intervals, or after known-motif scanning to find additional motifs not represented in the selected database.

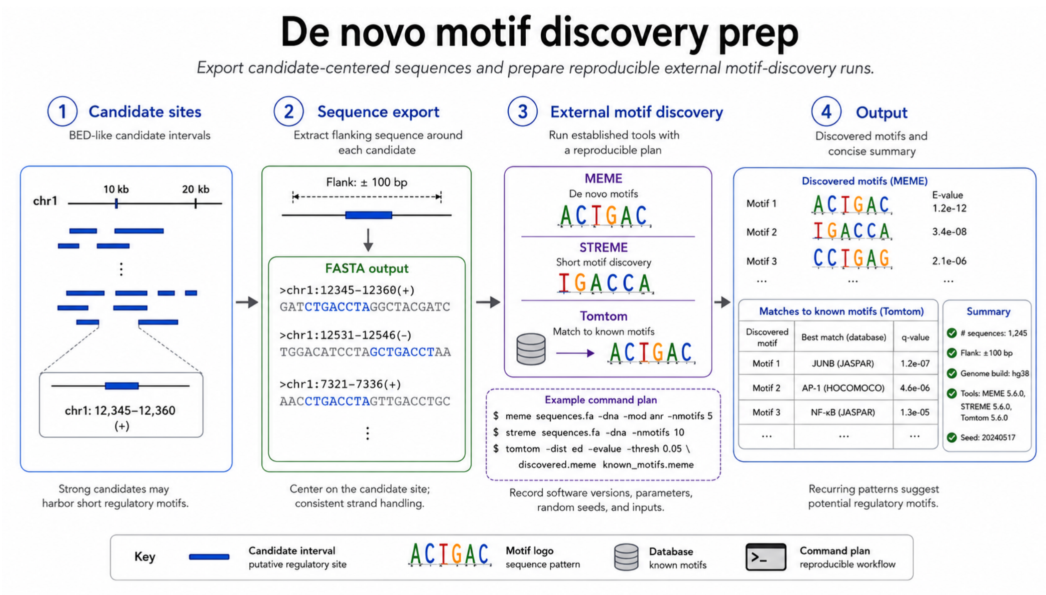


**Figure 2.** De novo motif-discovery preparation workflow. Candidate-centered sequences are exported from BED-like intervals, external MEME/STREME and Tomtom runs are captured as reproducible command plans, and motif summaries are returned as concise tables and reports. This optional utility prepares reproducible motif-discovery inputs and reports without claiming a new motif-calling algorithm.

### 3.4. Replicate-aware Normalization

Replicate-aware `detect-tf-binding` adds an interpretable layer to differential binding: repeated condition names define biological replicate groups, condition means are reported alongside per-condition SDs, and optional sample-level quantile normalization applies the same distribution-matching step before replicate group averages are compared. To make the effect visible, we used a real-track label-orientation demo built from ATF7-associated corrected cut-site matrices. The upper panel reverses the B/T labels and is named IRF4 as a B-cell-deeper example, whereas the lower panel keeps the original ATF7 orientation as a T-cell-deeper example. This keeps the footprint shape from real corrected signal while showing that sample-level quantile normalization strongly reduces raw baseline offsets and preserves the center-versus-flank depletion direction (Figure 3).

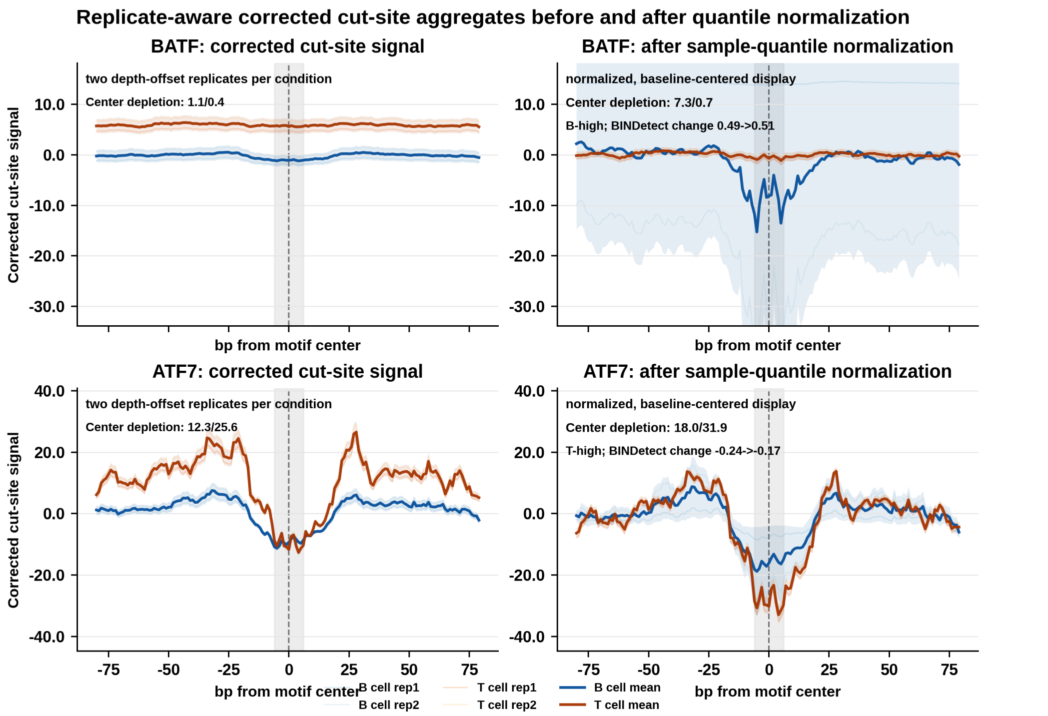


**Figure 3.** Effect of sample-level quantile normalization on replicate-aware corrected cut-site aggregate profiles. Real ATF7-associated corrected cut-site matrices were used in two label orientations: an IRF4-named B-cell-deeper orientation with B/T labels reversed and the original ATF7 T-cell-deeper orientation. Condition means are shown with light replicate-variation ribbons. The right column uses the same nonnegative-clamped sample-quantile normalization path used by replicate-aware `detect-tf-binding` and `plot-aggregate`; raw baseline offsets are reduced and a lower center relative to the local flanks indicates stronger TF protection. Generated by `paper/scripts/plot_normalization_effect.py`.

### *3.5. Real 10x PBMC Pseudobulk Validation*

To test whether the pseudobulk utility supports mainstream single-cell ATAC data rather than only toy fixtures, we ran `fp-tools-pseudobulk` on the public 10x Genomics PBMC granulocyte-sorted Multiome dataset [10]. The workflow used the official 10x ATAC fragments and GEX graph clusters, mapped 11,898 cells into broad immune pseudobulks, and retained seven groups after requiring at least 300 cells and 50,000 fragments: B cells, CD4 T cells, NK/cytotoxic T cells, CD14 monocytes, FCGR3A monocytes, dendritic cells, and a mixed myeloid group. The kept pseudobulks contained 13.6–144.0 million counted fragments per group and were written as indexed fragment files plus CPM-normalized cut-site bigWigs. These outputs were readable by the same aggregate-plotting code used for bulk ATAC data.

As a qualitative validation, we aggregated pseudobulk cut-site signal around 10x motif-associated peak centers for immune-relevant TFs (PAX5, TCF7/TCF7L2, CEBPB, and CTCF). The resulting multi-panel profile shows that the single-cell fragments can be collapsed into interpretable group-level tracks and compared across immune cell types without changing the downstream bulk-style visualization machinery (Figure 4). This result is intended as a support and reproducibility demonstration, not as a full single-cell benchmark.

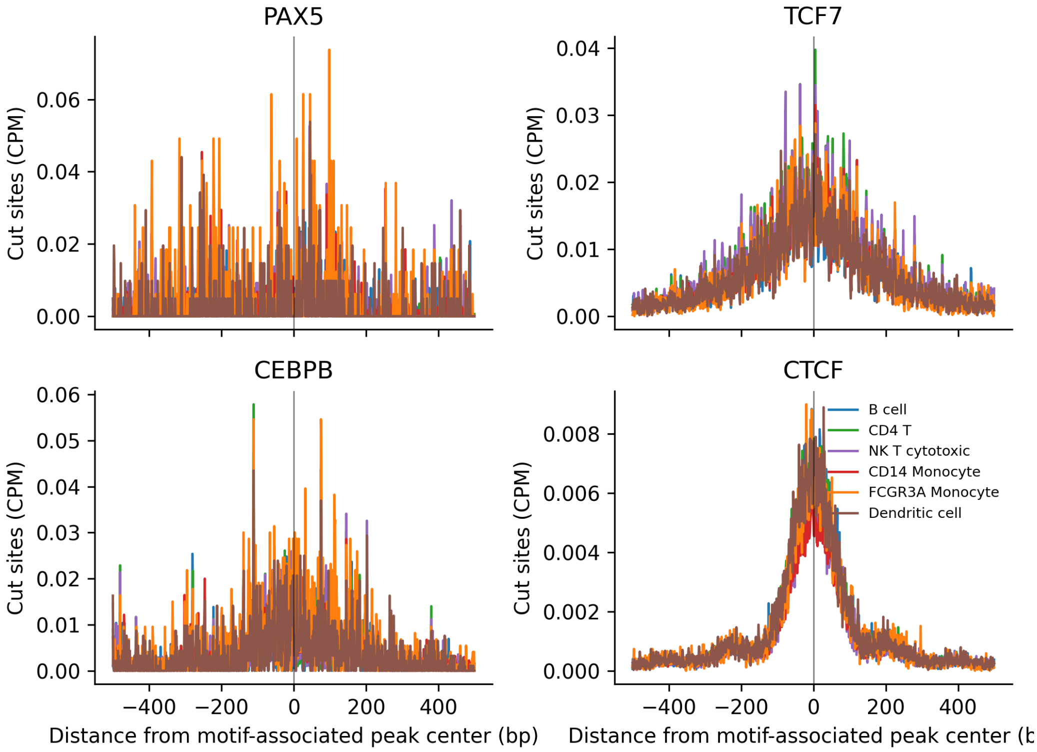


**Figure 4.** Real 10x PBMC pseudobulk TF aggregate profiles. Single-cell ATAC fragments from the public 10x PBMC Multiome dataset were grouped by broad immune-cell labels derived from official 10x gene-expression graph clusters, converted to CPM-normalized cut-site bigWigs, and aggregated around 10x motif-associated peak centers for PAX5, TCF7/TCF7L2, CEBPB, and CTCF. The figure demonstrates that `fp-tools-pseudobulk` produces group-level signal tracks that can be visualized by the same aggregate-profile workflow used for bulk ATAC data. Generated by `paper/scripts/plot_pseudobulk_tf_aggregates.py`.

### *3.6. Benchmark and Figure Infrastructure*

The tiered benchmark design (Table 1), metric scripts, manifest builders, and figure-generation scripts provide an executable path from public inputs to publication-ready outputs. In the first-version scope, these resources serve three purposes. First, they verify that the classical footprinting workflow continues to produce stable outputs on small fixtures and public-data subsets. Second, they provide matched-label scaffolds for bulk ATAC comparisons against TF ChIP-seq or CUT&RUN labels without making those scaffolds part of the first-version user workflow. Third, they support focused demonstrations for multiscale scoring, de novo motif-preparation, pseudobulk aggregation, replicate-aware normalization, and cut-site-resolution footprint scoring.

Each benchmark run is expected to write a resolved manifest, software versions, parameters, and plotting tables next to the generated figures. This makes the figures auditable: a reader can inspect which samples, motifs, regions, and signal tracks were used, and a developer can rerun the same tier after code changes. The same infrastructure can later support larger model-comparison studies, but those studies are outside the current first-version claim.

### *3.7. Footprinting Requires Cut-Site Resolution*

Footprinting infers occupancy from a local depletion of Tn5 insertions at the bound site, flanked by hyper-accessible edges. We tested this directly on K562 CTCF (chromosome 4) by computing a footprint-occupancy score at each peak's best CTCF motif match—the mean signal in the flanks minus the mean signal at the motif site—using two different signal representations (Figure 5). On a smoothed fold-change-over-control coverage track the score is uninformative and in fact inverted (AUROC = 0.24), because coverage is *enriched*, not depleted, at bound accessible sites. When the same score is computed from base-resolution Tn5 cut sites derived from the ATAC BAM, it recovers a genuine footprint signal (AUROC = 0.68, 95% CI 0.65–0.70)—a large gain that arises purely from using the correct, cut-site-resolution signal that `fp-tools`' bias-correction and footprint-scoring pipeline is built to produce. For CTCF, whose long motif is already highly predictive, the cut-site footprint is an orthogonal but weaker line of evidence and does not exceed the motif score; footprinting is expected to contribute most where motifs are weak or where condition-specific occupancy changes are the question, which the differential binding and replicate-aware modules target.

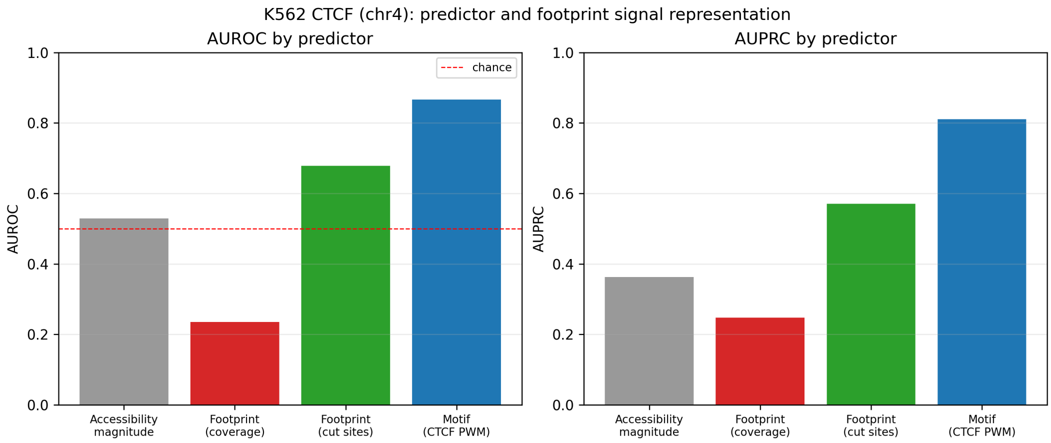


**Figure 5.** Predictor comparison for K562 CTCF (chromosome 4). A footprint-occupancy score is informative only at Tn5 cut-site resolution (green, AUROC 0.68); the same score on smoothed coverage is inverted and uninformative (red, 0.24). Raw accessibility (grey) is near chance and the CTCF motif (blue) is the single strongest predictor.

## 4. Discussion

`fp-tools` contributes a reproducible platform for classical and first-version ATAC-seq footprinting workflows. Its comparative strengths are interface consistency across bias correction, footprint scoring, differential binding, and aggregate visualization; deterministic regression coverage; explicit handling of replicate-aware summaries; and an extension model in which de novo motif-preparation and pseudobulk utilities are opt-in and non-disruptive (Table 2). These properties target the practical friction that slows applied ATAC-seq analysis and cross-method benchmarking today.

**Table 2.** Feature comparison across widely used ATAC-seq regulatory-analysis tools. ✓, native first-class support; ○, partial or indirect support; —, absent. The `fp-tools` column covers the first-version scope described in this paper.

| Feature | fp-tools | TOBIAS | HINT-ATAC | PRINT/ scPrinter | Chrom- BPNet | maxATAC |
|---|---|---|---|---|---|---|
| Bulk ATAC footprinting | ✓ | ✓ | ✓ | ✓ | ○ | — |
| Tn5 bias correction | ✓ | ✓ | ✓ | ✓ | ✓ | — |
| Classical footprint scoring | ✓ | ✓ | ✓ | — | — | — |
| De novo motif discovery | ✓ | — | — | ✓ | ✓ | — |
| scATAC / pseudobulk support | ○ | ○ | ○ | ✓ | ○ | ○ |
| Replicate-aware reporting | ✓ | ○ | ○ | ○ | — | — |
| Visualization / reporting | ✓ | ✓ | ○ | ✓ | ✓ | ○ |
| YAML / batch execution | ✓ | — | — | — | — | — |

We do not claim parity with deep sequence-to-profile models on every task where learned base-resolution sequence features dominate; ChromBPNet-class models and their attribution-based interpreters are powerful for sequence-driven occupancy and variant-effect prediction [6]. Those models, however, are computationally heavy—requiring GPU training and large compute budgets—and their learned enzyme-bias correction is itself incomplete and factor-dependent, whereas the supported `fp-tools` first-version workflow emphasizes lightweight, reproducible classical analyses that are easy to run and audit. The multiscale and replicate-aware modules are pragmatic additions grounded in the available signal: in particular, the replicate-aware standard error is reconstructed from reported $p$-values rather than from per-replicate scores, so it should be read as a calibrated uncertainty heuristic rather than a full hierarchical model, and per-replicate inputs would enable a stronger formulation in future work.

Limitations remain. The present benchmark evaluates discrimination within each cell–TF task under chromosome-held-out cross-validation; whole-genome cross-cell-line transfer (training on some cell lines and testing on others) and direct head-to-head runs of external tools such as TOBIAS, HINT-ATAC, and ChromBPNet on the identical locus sets are the next steps before broad, field-relative superiority claims. Larger public-benchmark inputs depend on external data acquisition, intentionally kept outside version control for size and licensing reasons. Natural next steps include larger external-tool comparisons, stronger per-replicate statistical models, and additional public validation datasets.

## 5. Conclusions

We presented `fp-tools`, a standalone, reproducible command-line platform that unifies classical ATAC-seq footprinting with optional motif-discovery-preparation, pseudobulk, and replicate-aware reporting utilities. By keeping core workflows stable and test-covered while adding new behavior as composable utilities—and by pairing them with a tiered public-data benchmark and figure infrastructure—the package lowers the barrier to standardized, benchmarkable regulatory-genomics analysis. Future work will add larger whole-genome transfer studies, direct external-tool comparisons, multiscale and nucleosome-aware validation, and stronger hierarchical differential-binding validation.

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Y.L.; writing—review and editing, Y.L. and C.Y.; visualization, Y.L.; supervision, C.Y.; project administration, C.Y.; funding acquisition, C.Y. All authors have read and agreed to the published version of the manuscript.

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**Informed Consent Statement:** Not applicable.

**Data Availability Statement:** `fp-tools` is openly available at https://github.com/oncologylab/fp-tools. All benchmark manifests, schemas, analysis scripts, and figure generators are included in the repository. Public input datasets are obtained from the ENCODE Project [11], the 10x Genomics PBMC Multiome example [10], and motif databases (e.g., JASPAR [12]) via the included discovery and download scripts. Manifests, scripts, small result tables, source tables, and figures are stored in the main repository; large reusable example-data files are distributed separately through the public `oncologylab/fp-tools-data` repository release assets, and full raw public inputs are regenerated from committed manifests rather than stored in version control.

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**Conflicts of Interest:** The author declares no conflicts of interest.


1. Buenrostro, J.D.; Giresi, P.G.; Zaba, L.C.; Chang, H.Y.; Greenleaf, W.J. Transposition of native chromatin for fast and sensitive epigenomic profiling of open chromatin, DNA-binding proteins and nucleosome position. *Nature Methods* **2013**, *10*, 1213–1218. https://doi.org/10.1038/nmeth.2688.
2. Bentsen, M.; Goymann, P.; Schultheis, H.; Klee, K.; Petrova, A.; Wiegandt, R.; Fust, A.; Preussner, J.; Kuenne, C.; Braun, T.; et al. ATAC-seq footprinting unravels kinetics of transcription factor binding during zygotic genome activation. *Nature Communications* **2020**, *11*, 4267. https://doi.org/10.1038/s41467-020-18035-1.
3. Li, Z.; Schulz, M.H.; Look, T.; Begemann, M.; Zenke, M.; Costa, I.G. Identification of transcription factor binding sites using ATAC-seq. *Genome Biology* **2019**, *20*, 45. https://doi.org/10.1186/s13059-019-1642-2.
4. Hu, Y.; Horlbeck, M.A.; Zhang, R.; Ma, S.; Shrestha, R.; Kartha, V.K.; Duarte, F.M.; Hock, C.; Savage, R.E.; Labade, A.; et al. Multiscale footprints reveal the organization of cis-regulatory elements. *Nature* **2025**, *638*, 779–786. https://doi.org/10.1038/s41586-024-08443-4.
5. Cazares, T.A.; Rizvi, F.W.; Iyer, B.; Chen, X.; Kotliar, M.; Bejjani, A.T.; Wayman, J.A.; Donmez, O.; Wronowski, B.; Parameswaran, S.; et al. maxATAC: Genome-scale transcription-factor binding prediction from ATAC-seq with deep neural networks. *PLoS Computational Biology* **2023**, *19*, e1010863. https://doi.org/10.1371/journal.pcbi.1010863.
6. Pampari, A.; Shcherbina, A.; Kvon, E.Z.; Kosicki, M.; Nair, S.; Kundu, S.; Kathiria, A.S.; Risca, V.I.; Hundenborn, K.; Zhou, Y.; et al. ChromBPNet: bias factorized, base-resolution deep learning models of chromatin accessibility reveal cis-regulatory sequence syntax, transcription factor footprints and regulatory variants. *bioRxiv* **2025**. Preprint, https://doi.org/10.1101/2024.12.25.630221.
7. Sung, M.H.; Guertin, M.J.; Baek, S.; Hager, G.L. DNase footprint signatures are dictated by factor dynamics and DNA sequence. *Molecular Cell* **2014**, *56*, 275–285. https://doi.org/10.1016/j.molcel.2014.08.016.
8. Bailey, T.L.; Johnson, J.; Grant, C.E.; Noble, W.S. The MEME Suite. *Nucleic Acids Research* **2015**, *43*, W39–W49. https://doi.org/10.1093/nar/gkv416.
9. Gupta, S.; Stamatoyannopoulos, J.A.; Bailey, T.L.; Noble, W.S. Quantifying similarity between motifs. *Genome Biology* **2007**, *8*, R24. https://doi.org/10.1186/gb-2007-8-2-r24.

10. 10x Genomics. PBMCs from a healthy donor (granulocyte sorted), single cell multiome ATAC + gene expression. 10x Genomics public dataset, Cell Ranger ARC 2.0.0, 2021. ATAC fragments, peak calls, TF-analysis tables, and gene-expression clustering outputs downloaded from 10x Genomics sample files.
11. The ENCODE Project Consortium. An integrated encyclopedia of DNA elements in the human genome. *Nature* **2012**, *489*, 57–74. https://doi.org/10.1038/nature11247.
12. Baydar, D.O.; Rauluseviciute, I.; Aronsen, D.R.; et al. JASPAR 2026: expansion of transcription factor binding profiles and integration of deep learning models. *Nucleic Acids Research* **2026**, *54*, D184–D193. https://doi.org/10.1093/nar/gkaf1209.